Abed Zahed

RF & Wireless Systems Engineer | Telecom & Wireless Sensing | Engineering Lead

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Nationality: Swedish | Location: Sharjah, UAE

EXPERIENCE

Personal Project: Job Search Automation Platform

Independent

07/2026 - Present 📌 *Sharjah, UAE (Remote)*

- Built an end-to-end Python pipeline with a FastAPI REST (Representational State Transfer) backend (SQLAlchemy/Postgres, Celery async queue) scraping GCC (Gulf Cooperation Council)/EU job postings and generating AI (Artificial Intelligence)-tailored application materials as a multi-user SaaS (Software as a Service) product, plus a Next.js frontend and Chrome extension client.
- Set up CI/CD (Continuous Integration/Continuous Deployment) via GitHub Actions and PowerShell-based automation orchestration integrating multiple Ollama Cloud LLM (Large Language Model) models into the delivery pipeline, including scripted task delegation, retry/failure-escalation logic, and result verification.

Academic Research Project: Satellite RF Link Budget & Modulation Simulation

American University of Sharjah (Research Collaboration)

06/2026 - Present 📌 *Sharjah, UAE*

- Designed a physics-based link budget model for a GEO S-band satellite link in an ongoing academic research collaboration with the American University of Sharjah; modeled multipath, scintillation, and Faraday rotation referencing ITU-R M.1343 and ECSS-E-ST-50-05C, targeting peer-reviewed publication.
- Implemented CCSDS LDPC adaptive coding and modulation in MATLAB, spanning 3 active QPSK profiles (r=1/2 to r=3/4) and extending evaluation to 8PSK r=7/8 and DVB-S2 16APSK per ETSI EN 302 307-1 for robust link performance across varying channel conditions.
- Ran Monte Carlo simulations generating BER waterfall curves via MATLAB Communications Toolbox; achieved 9.5 dB coding gain vs uncoded QPSK, reducing transmit power by 9x.

SUMMARY

RF and wireless systems engineer with 5+ years at Ericsson (led MRC/MRCx implementation, 2.7 dB signal gain; MATLAB, Python, and FPGA test bench for SISO/XPIC/MIMO) and Veoneer (65% to 98% production turnaround via root cause analysis). M.Sc. Wireless, Photonics and Space Engineering, Chalmers; B.Sc. Electrical Engineering, AUS; 4 IEEE publications in RF sensing.

LANGUAGES

Arabic	●●●●●	Native
English	●●●●○	Full Professional
Swedish	●●●○○	Elementary (improving)
Spanish	●●○○○	Elementary

LICENSES

✓ EU (Sweden) Driving License: Category B

✓ UAE Driving License

Consultant Engineer

Fun Robotics

11/2024 - 08/2025 📌 *Dubai, UAE*

- Managed end-to-end preparation and implementation of technical training courses, coordinating between Fun Robotics and TDRA (UAE Telecom Regulatory Authority) on logistics, scheduling, and program delivery.
- Designed and delivered technical workshops with TDRA, managing curriculum development and coordination with Etisalat Academy.
- Led a team to develop and deliver technical training on wireless communications and RF systems.

Tech Consultant

Mpya Sci & Tech

10/2020 - 08/2024 📌 *Gothenburg, Sweden*

- Worked as engineering consultant across Ericsson and Veoneer client sites via Mpya Sci & Tech's professional-services model.
- Held After-Work Committee Coordinator and Consultant Buddy roles at Mpya, driving team culture initiatives and onboarding new consultants.

Microwave DSP Config Developer (Consultant)

Ericsson

03/2022 - 12/2023 📌 *Gothenburg, Sweden*

- Implemented MRC and novel MRCx combining on XPIC channel modes for Ericsson MiniLink; achieved 2.7 dB signal power gain.
- Extended modulation profile work beyond MRC/MRCx combining to cover higher-order QAM schemes up to 32-QAM.
- Developed MATLAB and Python test bench for MiniLink validating SISO, XPIC, and MIMO channel configurations end-to-end.

Process Engineer (Consultant)

Veoneer / Magna International

11/2020 - 01/2022 📌 *Vårgårda, Sweden*

- Improved machine performance from 65% to 98% by leading a team of 3 optics engineers in troubleshooting, root cause analysis, and documentation, part of the camera production line's industrialization and ramp-up to full-scale manufacturing.
- Directed incident and problem resolution for camera production; coordinated cross-functional investigations with production, engineering, and maintenance teams.
- Built Power BI dashboards and Python pipelines to track production efficiency and process capability KPIs; adopted org-wide for yield monitoring.

SKILLS

CORE COMPETENCIES

- Technical Training & Knowledge Transfer
- Regulatory Compliance (ETSI / ANSI / IEC)
- Cross-functional Team Leadership
- TDRA / UAE Telecom Regulatory Coordination

TELECOM & ENGINEERING

- Digital Modulation and Coding (QPSK, 8PSK, DVB-S2, CCSDS LDPC)
- Wireless & Microwave Systems (LTE/5G, Radio Links)
- RF Testing, Validation & Performance Analysis
- RF Lab Equipment and Instrumentation (VNA, Spectrum Analyzer, Signal Generators, Network Analyzers)
- Circuit and PCB Design (RF antenna PCBs, fractal antennas, diagnostic test setups; Altium)
- RF Front-End Systems (LNAs, PAs, mixers, filters, T/R modules; systems-level design)
- Digital Signal Processing (DSP)
- Machine Learning (ANN, SVM , RF sensing classification)
- MIMO / XPIC / SISO channel modeling
- Antenna Prototyping (patch antennas, fractal antennas, fabrication and measurement)

TOOLS & PLATFORMS

- Electronics Testing and Diagnostics (RF systems, channel emulation, performance validation)
- FPGA-based channel emulation
- MATLAB
- Python (test automation, data analysis)
- Simulink
- HFSS
- ADS, CST Microwave Studio (EM simulation)
- Power BI

Space Systems Research (M.Sc. Program)

Onsala Space Observatory / Chalmers University

2018 - 2019 📌 *Onsala, Sweden*

- Completed research training at Onsala Space Observatory covering satellite communications, link budgets, and space systems engineering as part of M.Sc. in Wireless, Photonics and Space Engineering at Chalmers.
- Studied orbital mechanics and satellite platform design, radar systems with Doppler processing and FMCW waveforms, and satellite positioning/GNSS with pseudorange-based differential positioning.
- Operated GNSS tracking, satellite positioning, and antenna dish control systems in an active deep space observatory, generating the raw datasets used for deep-space data processing and analysis.

Master's Thesis: RF Researcher

RISE Research Institutes of Sweden

02/2019 - 06/2019 📌 *Borås / Gothenburg, Sweden*

- Increased detection sensitivity by 3x via patch antenna research for transformer partial discharge monitoring.
- Designed and tested a prototype patch antenna using VNA, spectrum analyzer, and signal generators; calibrated test setup demonstrated performance comparable to standardized measurement setups.
- Measured and characterized radiation patterns for the patch antenna prototype; validated gain, return loss, and 50Ω impedance matching across the UHF frequency range.

Robotics Engineer

Fun Robotics

06/2016 - 08/2017 📌 *Dubai, UAE*

- Taught wireless communication fundamentals, protocol configuration, embedded systems, and RF signal concepts to students age 7-20.
- Designed comprehensive teaching curricula and supervised student project work across robotics and software platforms for school- and university-level students; adopted as educational resources for the academic industry.
- Designed and manufactured robotic solutions as part of the product development team.

KEY ACHIEVEMENTS

♦ MRC/MRCx Implementation

Led implementation on Ericsson MiniLink products achieving 2.7 dB gain in received signal power, effectively doubling received power.

♦ UAE Telecom Training Delivery

Designed and delivered technical training with UAE TDRA in partnership with Etisalat Academy, supporting regulatory capability building.

♦ Production Performance Turnaround

Improved camera production line performance from 65% to 98% at Veoneer/Magna by leading a team of 3 optics engineers through root cause analysis and corrective action.

♦ RF Sensing Research

Increased detection sensitivity 3x via patch antenna research at RISE Sweden; co-authored 4 IEEE publications in RF sensing and diagnostic antenna design.

♦ Operational Analytics

Developed Power BI dashboards adopted org-wide at Veoneer for production yield monitoring and Python-automated data analysis pipelines.

Visiting Researcher

American University of Sharjah

01/2015 - 10/2015 📌 *Sharjah, UAE*

- Co-authored 4 IEEE journal and conference publications in RF sensing and diagnostic antenna design.
- Designed PCB-based RF antennas with 50Ω coaxial feeds; applied transmission line theory for impedance matching and optimized RF circuit feed networks.
- Simulated 4th-order Hilbert fractal antenna designs in HFSS; characterized 2D and 3D radiation patterns to verify semi-spherical coverage and directivity at VHF/UHF (0.3-6 GHz).

EDUCATION

M.Sc. in Wireless, Photonics, and Space Engineering

Chalmers University of Technology

2017-2019 📌 *Gothenburg, Sweden*

B.Sc. in Electrical Engineering (Minor: Computer Engineering)

American University of Sharjah

2011-2015 📌 *Sharjah, UAE*

Includes CCNA (Cisco Certified Network Associate) introductory coursework, familiar with Cisco Packet Tracer simulation tools